Unit-2: Descriptive Analytics

Topic: Introduction to Descriptive Analytics

Descriptive analytics is the **first level of data analysis** in data science and business intelligence. Before building predictions or machine learning models, organizations must first **understand the data they already have**.

1. What is Descriptive Analytics

Definition: Descriptive Analytics is the process of analyzing historical data to summarize and understand what has happened in the past.

It focuses on:

- organizing data
- summarizing information
- identifying patterns or trends

Simple idea: **Descriptive Analytics = Understanding past data.**

2. Purpose of Descriptive Analytics

The main objective is to **convert raw data into meaningful information**.

Organizations use descriptive analytics to:

Purpose	Explanation
Understand past performance	analyze historical data
Identify trends	observe patterns over time
Monitor operations	track business activities
Support decision making	provide factual insights

3. Example of Descriptive Analytics

Example: Retail Store

A retail store analyzes last year's sales data.

Month	Sales (₹)
January	50,000
February	60,000
March	55,000

From this data we can determine:

- average monthly sales
- highest sales month
- overall sales trend

This is **descriptive analytics**.

4. Role in Data Science

Descriptive analytics is the **first step in the analytics hierarchy**.

Stage	Question Answered
Descriptive Analytics	What happened?
Diagnostic Analytics	Why did it happen?
Predictive Analytics	What will happen?
Prescriptive Analytics	What should we do?

Without descriptive analytics, the other stages cannot function properly.

5. Techniques Used in Descriptive Analytics

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Descriptive analytics uses statistical techniques to summarize data.

Technique	Purpose
Mean	average value
Median	middle value
Mode	most frequent value
Range	difference between highest and lowest
Standard Deviation	measure data variability
Frequency Distribution	count of occurrences

These methods help simplify large datasets.

6. Data Visualization in Descriptive Analytics

Visualization helps interpret data easily.

Common charts used:

Chart	Purpose
Bar Chart	compare categories
Line Graph	show trends over time
Pie Chart	show proportions
Histogram	show distribution of data

Example:

A line graph may show **monthly website visitors**.

7. Process of Descriptive Analytics

Descriptive analytics follows a structured process.

Data Collection

↓

Data Cleaning

↓

Data Organization

↓

Statistical Analysis

↓

Visualization & Reports

This process converts **raw data into understandable insights**.

8. Real-World Applications

Descriptive analytics is used in many industries.

Industry	Example
Retail	monthly sales reports
Healthcare	patient admission statistics
Banking	transaction summaries
Marketing	campaign performance analysis
Education	student performance analysis

9. Example: Website Traffic Analysis

A website records daily visitors.

Day	Visitors
Monday	1,200
Tuesday	1,450

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Wednesday	1,300

Descriptive analytics helps determine:

- average daily visitors
- peak traffic day
- overall traffic trend

10. Advantages of Descriptive Analytics

Advantage	Explanation
Easy to understand	simple statistical methods
Quick insights	summarize large data quickly
Foundation for advanced analytics	prepares data for predictive models

11. Limitations

Descriptive analytics has some limitations.

Limitation	Explanation
Cannot explain causes	only describes past events
Cannot predict future	no forecasting ability
Limited decision support	deeper analysis required

12. Example of Descriptive Analytics in Data Science

Suppose a company collects customer data:

Customer	Purchases
A	5
B	8
C	3

Using descriptive analytics we calculate:

- average purchases
- most active customer
- purchase distribution

This gives **basic insights about customer behavior**.

Final Concept Summary

Concept	Meaning
Descriptive Analytics	analyzing historical data
Main Question	What happened?
Techniques	statistical summaries and visualization
Purpose	understand patterns and trends

Core Idea

Raw Data → Summarization → Insights

Descriptive analytics transforms **large datasets into understandable summaries**, forming the **foundation for deeper statistical and machine learning analysis**.

Topic: Data Types

Understanding **data types** is fundamental in descriptive analytics because the **type of data determines which statistical methods and visualizations can be used**. Before analyzing any dataset, data scientists must identify the correct data type.

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1. What are Data Types

Definition: Data types refer to the classification of data based on the type of values they represent and the operations that can be performed on them.

In statistics and data science, data is mainly classified into **two major categories**:

Main Category	Description
Qualitative Data	descriptive data representing categories
Quantitative Data	numerical data representing measurable quantities

2. Qualitative Data (Categorical Data)

Definition: Qualitative data represents characteristics or categories that cannot be measured numerically.

It describes **qualities or attributes** rather than numbers.

Examples

Example	Description
Gender	Male / Female
Color	Red, Blue, Green
City	Ahmedabad, Mumbai
Product Category	Electronics, Clothing

Even if numbers are used (e.g., 1 = Male, 2 = Female), they represent **categories**, not numerical values.

Types of Qualitative Data

Qualitative data is divided into two types:

Type	Description
Nominal Data	categories with no specific order
Ordinal Data	categories with a meaningful order

Nominal Data: Nominal data represents **labels or names without any ranking**.

Examples:

Data	Category
Blood Group	A, B, AB, O
Nationality	Indian, American
Product Type	Mobile, Laptop

No category is greater or smaller than another.

Ordinal Data: Ordinal data represents **categories with a meaningful order or ranking**.

Examples:

Rating	Meaning
1 Star	Poor
2 Stars	Fair
3 Stars	Good
4 Stars	Very Good
5 Stars	Excellent

Although there is order, the **difference between ranks is not necessarily equal**.

3. Quantitative Data (Numerical Data)

Definition: Quantitative data represents numerical values that can be measured or counted.

These values allow **mathematical calculations**.

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Examples include:

- height
- weight
- age
- salary
- number of products sold

Types of Quantitative Data

Quantitative data has two subtypes.

Type	Description
Discrete Data	countable values
Continuous Data	measurable values

Discrete Data: Discrete data represents **countable values that cannot be divided further**.

Examples:

Example	Value
Number of students in a class	40
Number of cars in parking lot	25
Number of products sold	120

Values are typically **whole numbers**.

You cannot have **2.5 students**.

Continuous Data: Continuous data represents **measurable values that can take any value within a range**.

Examples:

Example	Possible Values
Height	170.2 cm
Temperature	36.5 °C
Weight	65.7 kg

Continuous data may include **decimal values**.

4. Summary of Data Types

Data Category	Subtype	Example
Qualitative	Nominal	gender, blood group
Qualitative	Ordinal	ratings, rankings
Quantitative	Discrete	number of students
Quantitative	Continuous	height, temperature

5. Why Data Types are Important

Data type determines:

- which statistical methods to use
- which visualizations are appropriate
- how data should be processed

Example

Data Type	Best Visualization
Nominal	bar chart
Ordinal	bar chart

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Discrete	histogram
Continuous	line graph

Correct identification of data type ensures **accurate analysis**.

6. Example Dataset

Consider the following dataset of customers:

Customer	Age	Gender	Rating
A	25	Male	4
B	30	Female	5

Data types:

Variable	Data Type
Age	Quantitative (Continuous)
Gender	Qualitative (Nominal)
Rating	Qualitative (Ordinal)

7. Role of Data Types in Data Science

Before applying statistical analysis or machine learning models, data scientists must:

1. identify data types
2. convert them into suitable formats
3. select appropriate analytical techniques

Example: Machine learning models treat **numerical and categorical variables differently**.

Final Concept Summary

Data Type	Meaning
Qualitative	categorical descriptive data
Quantitative	numerical measurable data
Discrete	countable numbers
Continuous	measurable values

Core Idea

```
Data
 ├── Qualitative
 │    ├── Nominal
 │    └── Ordinal
 │
 └── Quantitative
      ├── Discrete
      └── Continuous
```

Understanding data types is the **foundation for all statistical analysis in descriptive analytics**.

Before studying **measurement scales**, it is important to clearly understand the four common data types used in descriptive analytics:

- Nominal
- Ordinal
- Discrete
- Continuous

These belong to the two main groups of data:

Main Category	Data Type

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Qualitative (Categorical)	Nominal, Ordinal
Quantitative (Numerical)	Discrete, Continuous

1. Nominal Data

Definition: Nominal data represents categories or labels that have no natural order or ranking.

The word *nominal* comes from the word **"name"**, meaning the data simply names or labels a category.

Characteristics

Property	Explanation
No order	categories cannot be ranked
Non-numerical meaning	numbers (if used) are just labels
Only classification	used to group items

Examples

Example	Categories
Gender	Male, Female
Blood group	A, B, AB, O
Nationality	Indian, American, Japanese
Car brand	Toyota, Honda, BMW

Example dataset:

Person	Blood Group
A	O
B	AB
C	B

There is **no ranking** between these categories.

Mathematical Operations

Nominal data allows only:

- counting
- frequency calculation

Example:

Blood Group	Frequency
A	10
B	8
O	12

But we **cannot calculate averages**.

2. Ordinal Data

Definition: Ordinal data represents categories that have a meaningful order or ranking.

However, the **difference between categories is not measurable**.

Characteristics

Property	Explanation
Ordered categories	ranking exists
Unequal intervals	difference between ranks unknown
Comparative analysis	higher or lower ranking possible

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Examples

Example 1: Customer Satisfaction

Rating	Meaning
1	Very Poor
2	Poor
3	Average
4	Good
5	Excellent

The ranking is clear, but the difference between 3 and 4 is not exactly measurable.

Example 2: Education Level

Level
High School
Bachelor's Degree
Master's Degree
PhD

These categories are ordered but **not numerically measurable**.

Mathematical Operations

Ordinal data supports:

- comparison (>, <)
- ranking
- median calculation

But not precise mathematical operations like **mean**.

3. Discrete Data

Definition : Discrete data represents countable numerical values that can take only specific separate values.

These values are usually **whole numbers**.

Characteristics

Property	Explanation
Countable values	exact numbers
No fractions	usually integers
Finite possibilities	limited values

Examples

Example	Value
Number of students in class	45
Number of cars in parking lot	20
Number of books sold	350

Example dataset:

Day	Orders
Monday	120
Tuesday	135
Wednesday	110

You cannot have **120.5 orders**.

Mathematical Operations

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Discrete data supports full mathematical operations:

- addition
- subtraction
- mean
- variance
- probability calculations

4. Continuous Data

Definition :Continuous data represents measurable quantities that can take any value within a range.

These values can include **decimals and fractions**.

Characteristics

Property	Explanation
Measurable values	obtained through measurement
Infinite possibilities	many possible values
Decimal values allowed	fractions possible

Examples

Example	Value
Height	175.4 cm
Weight	68.7 kg
Temperature	36.5°C
Time	2.45 seconds

Example dataset:

Person	Height (cm)
A	170.5
B	165.3
C	178.9

Values can take **any number within a range**.

Mathematical Operations

Continuous data allows full statistical analysis:

- mean
- standard deviation
- regression analysis
- probability distributions

Key Differences

Feature	Nominal	Ordinal	Discrete	Continuous
Data Type	Qualitative	Qualitative	Quantitative	Quantitative
Order	No	Yes	Yes	Yes
Numeric Meaning	No	Limited	Yes	Yes
Example	Gender	Rankings	Number of students	Height

Simple Visualization

Data

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```
├── Qualitative
│     ├── Nominal
│     └── Ordinal
│
└── Quantitative
      ├── Discrete
      └── Continuous
```

Quick Real-Life Example
Consider a dataset of students:

Student	Gender	Rank	Age	Height
B	Female	2	21	168.2

Variable	Data Type
Gender	Nominal
Rank	Ordinal
Age	Discrete
Height	Continuous

Key Understanding

Data Type	Main Idea
Nominal	classification
Ordinal	ranking
Discrete	counting
Continuous	measurement

Topic: Types of Data Measurement Scales

In statistics and data science, **measurement scales** describe **how data is measured and what type of mathematical operations can be performed on that data**.
Different types of data allow **different levels of analysis**. Understanding measurement scales helps data scientists choose the correct statistical techniques.

1. What is a Measurement Scale
Definition: A **measurement scale** is a system used to classify and measure data according to its characteristics and mathematical properties.
Measurement scales define:

- how data values are interpreted
- whether data can be ordered
- whether arithmetic operations are meaningful

2. Four Types of Measurement Scales
There are **four main measurement scales in statistics**.

Scale	Level of Measurement	Example
Nominal Scale	classification	gender, blood group
Ordinal Scale	ranking	customer satisfaction rating
Interval Scale	equal intervals but no true zero	temperature in °C
Ratio Scale	equal intervals with true zero	height, weight

These scales represent **increasing levels of measurement accuracy**.

3. Nominal Scale
Definition: A **nominal scale** is the simplest measurement scale where data is classified into categories without any order or ranking.

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The word **nominal** means “name”.

Characteristics

Property	Explanation
Categories only	used for labeling
No ranking	categories cannot be ordered
No arithmetic operations	numbers have no mathematical meaning

Examples

Example	Categories
Gender	Male, Female
Blood Group	A, B, AB, O
Eye Color	Blue, Brown, Black
Country	India, USA, Japan

Even if numbers are used (1 = Male, 2 = Female), they are **just labels**.

Allowed Operations

For nominal data we can perform:

- counting
- frequency distribution
- mode

Example:

Blood Group	Frequency
A	15
B	10
O	20

4. Ordinal Scale

Definition: An **ordinal scale** represents categories that have a **meaningful order or ranking**. However, the **difference between ranks is not measurable**.

Characteristics

Property	Explanation
Ordered categories	ranking exists
Unequal intervals	difference between ranks not known
Relative comparison	higher or lower ranking possible

Examples

Example 1: Customer Satisfaction

Rating	Meaning
1	Very Poor
2	Poor
3	Average
4	Good
5	Excellent

Example interpretation:

- 4 is better than 3
- but the difference between 4 and 3 is not exactly measurable.

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Example 2: Education Level

Level
High School
Bachelor's Degree
Master's Degree
PhD

These categories follow a **logical order**.

Allowed Operations

For ordinal data we can perform:

- ranking
- comparison
- median calculation

But **mean calculation is not meaningful**.

5. Interval Scale

Definition: An **interval scale** is a measurement scale where the **difference between values is meaningful and equal**, but there is **no true zero point**.

Characteristics

Property	Explanation
Ordered data	values have ranking
Equal intervals	consistent difference between values
No absolute zero	zero does not mean absence

Examples

Temperature in Celsius

Temperature	Meaning
10°C	cool
20°C	warmer
30°C	hot

Difference between 10°C and 20°C is the same as between 20°C and 30°C.

However:

0°C does **not mean no temperature**.

Calendar Years

Year
2000
2010
2020

Difference between years is meaningful, but **year 0 does not represent absence of time**.

Allowed Operations

For interval data we can perform:

- addition
- subtraction
- mean calculation

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- standard deviation

But **ratios are not meaningful**.
Example: 30°C is **not twice as hot as 15°C**.

6. Ratio Scale
Definition: A **ratio scale** is the highest level of measurement. It has **all properties of interval scale plus a true zero point**.
True zero means **complete absence of the quantity**.

Characteristics

Property	Explanation
Ordered data	ranking exists
Equal intervals	consistent difference
True zero	zero means absence
Ratios meaningful	comparisons possible

Examples

Variable	Example
Height	170 cm
Weight	65 kg
Age	20 years
Income	₹50,000

If someone weighs **60 kg**, it is meaningful to say:
60 kg is **twice as heavy as 30 kg**.

Allowed Operations
Ratio scale allows all mathematical operations:

- addition
- subtraction
- multiplication
- division
- statistical analysis

This is the **most informative scale**.

7. Comparison of All Measurement Scales

Feature	Nominal	Ordinal	Interval	Ratio
Categories	Yes	Yes	Yes	Yes
Order	No	Yes	Yes	Yes
Equal intervals	No	No	Yes	Yes
True zero	No	No	No	Yes
Arithmetic operations	No	Limited	Partial	Full

8. Example Dataset
Consider a student dataset:

Student	Gender	Rank	Temperature	Height
A	Male	1	25°C	170 cm
B	Female	2	27°C	165 cm

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Variable	Measurement Scale
Gender	Nominal
Rank	Ordinal
Temperature	Interval
Height	Ratio

Final Concept Summary

Scale	Key Idea
Nominal	classification
Ordinal	ranking
Interval	equal intervals without true zero
Ratio	equal intervals with true zero

Simple Hierarchy
Nominal → Ordinal → Interval → Ratio
Each level provides **more information and more mathematical operations**.

Topic: Population and Sample

In statistics and data science, researchers often need to study **large groups of data**. Analyzing the entire dataset is not always possible due to **time, cost, and computational limitations**. Therefore, statisticians analyze a **smaller subset of the data**, called a sample.

1. Population
Definition: Population refers to the complete set of all individuals, objects, or observations that are of interest in a particular study.
It represents the **entire group being studied**.

Examples

Study	Population
Student performance	all students in a university
Election survey	all voters in a country
Product quality test	all manufactured products
Healthcare research	all patients with a specific disease

Example
Suppose a university wants to study the **average height of students**.
Population = **all students in the university**.
If the university has **20,000 students**, studying every student may be difficult.

Characteristics of Population

Feature	Description
Large size	may include thousands or millions of items
Complete dataset	includes all members of interest
Accurate results	if fully analyzed

However, studying the entire population is often **impractical**.

2. Types of Population
Population can be classified into two types.

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Type	Description
Finite Population	limited number of elements
Infinite Population	unlimited or extremely large number of elements

Finite Population Example

Example	Population Size
students in a class	40
employees in a company	500
houses in a city	fixed number

Infinite Population Example

Example
number of possible coin toss outcomes
number of possible measurements in physics experiments

These populations are **very large or theoretically unlimited**.

3. Sample

Definition: A **sample is a subset of the population selected for analysis**.
Instead of studying the entire population, researchers analyze a **representative group**.

Example

Population = 20,000 students in a university
Sample = 500 students selected randomly
By analyzing the sample, researchers estimate **population characteristics**.

Characteristics of a Good Sample

A good sample should be:

Property	Explanation
Representative	reflects population characteristics
Randomly selected	avoids bias
Adequate size	large enough for reliable results

4. Why Sampling is Used

Studying the entire population may not be practical.
Sampling is used because it:

Advantage	Explanation
Saves time	fewer observations required
Reduces cost	fewer resources needed
Faster analysis	smaller dataset
Practical	suitable for large populations

Example: Suppose a company produces **1 million smartphones**.
Testing every phone is impossible.
Instead:

- select **1000 phones as a sample**
- test their quality
- estimate overall product quality.

5. Population vs Sample

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Feature	Population	Sample
Size	entire dataset	smaller subset
Cost	expensive to analyze	cheaper
Time	time-consuming	faster
Accuracy	highly accurate	approximate estimation

6. Population Parameter and Sample Statistic

In statistics, measurements are defined differently for population and sample.

Concept	Meaning
Population Parameter	numerical characteristic of population
Sample Statistic	numerical characteristic of sample

Examples

Measure	Population Symbol	Sample Symbol
Mean	μ (mu)	$\bar{x}$ (x-bar)
Standard deviation	σ (sigma)	s
Proportion	p	$\hat{p}$

Example:

Population mean income = μ

Sample mean income = $\bar{x}$

Sample statistics are used to **estimate population parameters**.

7. Sampling Process

The sampling process usually follows these steps:

Define Population

↓

Select Sampling Method

↓

Choose Sample

↓

Collect Data

↓

Analyze Results

The results from the sample help infer characteristics of the population.

8. Example of Population and Sample in Data Science

Example: Online Streaming Platform

Population: All users of the platform.

Sample: 10,000 randomly selected users.

Data scientists analyze the sample to understand:

- viewing patterns
- preferred genres
- average watch time

These insights are then applied to the entire population.

9. Importance in Data Science

Population and sampling concepts are critical because:

- large datasets are common in big data systems

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- analyzing full data may be impractical
- sampling allows efficient analysis

Most machine learning models are trained using **sample datasets**.

Final Concept Summary

Concept	Meaning
Population	complete set of observations
Sample	subset of population
Parameter	measurement describing population
Statistic	measurement describing sample

Core Idea

Population → Select Sample → Analyze Sample → Estimate Population

Sampling enables researchers to **draw conclusions about large populations using smaller datasets**.

Percentile, Decile, and Quartile

These concepts are used in statistics to **divide a dataset into equal parts** so that we can understand **data distribution, ranking, and relative position of values**.

They are called **measures of position** because they show **where a value lies within a dataset**.

1. What are Measures of Position

Measures of position divide data into **equal sections**.

Measure	Number of Parts
Quartiles	4 equal parts
Deciles	10 equal parts
Percentiles	100 equal parts

These help us understand:

- ranking of values
- spread of data
- comparison between observations

2. Quartiles

Definition: Quartiles divide a dataset into four equal parts.

Each part contains **25% of the data**.

Types of Quartiles

Quartile	Symbol	Meaning
First Quartile	Q1	25% of data below this value
Second Quartile	Q2	median (50% of data below)
Third Quartile	Q3	75% of data below this value

Example Dataset

2, 4, 6, 8, 10, 12, 14, 16

Step 1: Find Median (Q2) :

Median = (8 + 10) / 2 = **9**

Step 2: Find Q1:

Lower half: 2, 4, 6, 8

Median = (4 + 6)/2 = **5**

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Step 3: Find Q3:
Upper half:
10, 12, 14, 16
Median = (12 + 14)/2 = **13**

Result

Quartile	Value
Q1	5
Q2	9
Q3	13

This means:

- 25% of values are below **5**
- 50% are below **9**
- 75% are below **13**

3. Deciles
Definition: Deciles divide a dataset into ten equal parts.
Each part contains **10% of the data**.

Decile Symbols

Decile	Meaning
D1	10% of data below
D2	20% of data below
D3	30% of data below
...	...
D9	90% of data below

Example
If a student's score lies in **D7**, it means:

- the score is higher than **70% of the students**.

Applications
Deciles are used in:

Field	Example
education	student ranking
finance	income distribution
marketing	customer segmentation

4. Percentiles
Definition: Percentiles divide a dataset into 100 equal parts.
Each part represents **1% of the data**.

Percentile Interpretation

Percentile	Meaning
P25	25% of data below
P50	median
P75	75% of data below
P90	90% of data below

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Example
Suppose a student scored in the **90th percentile**.
This means:

- the student performed better than **90% of students**.

Real-World Example
Standardized Exams
Many entrance exams report percentile scores.
Example:

Student Score	Percentile
85	92nd percentile

Meaning:
92% of students scored **below 85**.

5. Mathematical Formulas
Quartile Position
For ordered data:
Qk = k(n + 1) / 4
Where:

Symbol	Meaning
k	quartile number
n	number of observations

Decile Position
Dk = k(n + 1) / 10

Percentile Position
Pk = k(n + 1) / 100

6. Comparison of Quartiles, Deciles, and Percentiles

Measure	Number of Divisions	Percentage per Part
Quartiles	4	25%
Deciles	10	10%
Percentiles	100	1%

7. Visualization of Data Division
Quartiles
|----25%----|----25%----|----25%----|----25%----|

Deciles
|--10%--|--10%--|--10%--|--10%--|--10%--|--10%--|--10%--|--10%--|--10%--|--10%--|

Percentiles
|1%|1%|1%|...|1%| (100 divisions)

8. Importance in Data Science
These measures help in:

Application	Purpose
ranking	determine relative position

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data distribution	analyze spread of values
outlier detection	identify extreme values
comparison	compare different datasets

Example in Data Science
Suppose a company analyzes customer spending.

Percentile	Spending
25th percentile	₹200
50th percentile	₹500
75th percentile	₹900

Interpretation:

- 25% customers spend **less than ₹200**
- 50% spend **less than ₹500**

This helps identify **high-value customers**.

Final Concept Summary

Measure	Key Idea
Quartiles	divide data into 4 parts
Deciles	divide data into 10 parts
Percentiles	divide data into 100 parts

Core Idea
Dataset → Divide into Equal Parts → Understand Data Position
These measures help analysts **understand how values are distributed and ranked within a dataset**.

Measures of Central Tendency and Measures of Variation

When analyzing a dataset, two fundamental questions arise:

1. **Where is the center of the data?** → Measures of Central Tendency
2. **How spread out are the values?** → Measures of Variation (Dispersion)

Both concepts help summarize and understand data distribution.

Part 1: Measures of Central Tendency
1. Definition: Measures of central tendency are statistical measures that identify the central or typical value of a dataset.
They represent the **average location of data values**.

Main Types of Central Tendency

Measure	Meaning
Mean	arithmetic average
Median	middle value
Mode	most frequent value

1. Mean (Arithmetic Mean)
Definition: The **mean** is the sum of all values divided by the total number of observations.
Formula
Mean = (Sum of all observations) / (Number of observations)
Symbol:

Type	Symbol

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Population Mean	μ (mu)
Sample Mean	$\bar{x}$ (x-bar)

Example
Dataset:
5, 10, 15, 20
Step 1: Sum values
5 + 10 + 15 + 20 = **50**
Step 2: Divide by number of values
Mean = 50 / 4 = **12.5**

Properties of Mean

Property	Explanation
Uses all observations	every value contributes
sensitive to outliers	extreme values affect mean
most widely used	common statistical measure

2. Median
Definition: The **median** is the middle value of an ordered dataset.

Steps to Find Median

1. Arrange data in ascending order
2. Identify the middle value

Example 1 (Odd number of observations)
Dataset:
3, 5, 7, 9, 11
Median = **7**

Example 2 (Even number of observations)
Dataset:
4, 6, 8, 10
Median = (6 + 8) / 2 = **7**

Advantages

Advantage	Explanation
not affected by extreme values	robust measure
useful for skewed data	accurate central location

Example:
Income data often uses median instead of mean.

3. Mode
Definition: The **mode** is the value that appears **most frequently** in a dataset.

Example
Dataset:
2, 4, 4, 5, 7
Mode = **4**

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Types of Mode

Type	Description
Unimodal	one mode
Bimodal	two modes
Multimodal	multiple modes

Example
Dataset:
1, 2, 2, 3, 3
Modes = **2 and 3**

Comparison of Central Tendency Measures

Measure	Best Used When
Mean	data is symmetrical
Median	data contains outliers
Mode	categorical data

Part 2: Measures of Variation (Dispersion)
Central tendency shows the **center of data**, but it does not show **how spread out the values are**.
Measures of variation describe **how much the data values differ from each other**.

1. Range
Definition: The **range** is the difference between the maximum and minimum values in a dataset.
Formula
Range = Maximum value − Minimum value

Example
Dataset:
5, 8, 12, 20
Maximum = 20
Minimum = 5
Range = **20 − 5 = 15**

Limitation
Range depends only on **two values** and ignores other data points.

2. Variance
Definition: Variance measures how far each data value is from the mean.
It indicates the **degree of dispersion in the dataset**.

Formula
Population Variance
$\sigma^2 = \Sigma (xi - \mu)^2 / N$
Sample Variance
$s^2 = \Sigma (xi - \bar{x})^2 / (n - 1)$
Where:

Symbol	Meaning
xi	data value
μ	population mean

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$\bar{x}$	sample mean
N	population size
n	sample size

Example (Simplified)
Dataset:
4, 6, 8
Step 1: Find mean
Mean = (4 + 6 + 8) / 3 = **6**
Step 2: Calculate squared deviations

Value	Deviation	Square
4	-2	4
6	0	0
8	2	4

Step 3:
Variance = (4 + 0 + 4) / 3 = **2.67**

3. Standard Deviation
Definition: Standard deviation is the **square root of variance**.
It measures the **average distance of data values from the mean**.

Formula
Standard Deviation = √ Variance
Symbol:

Type	Symbol
Population	σ
Sample	s

Example
Variance = 2.67
Standard deviation =
√ 2.67 ≈ 1.63

Interpretation

Value	Meaning
Low standard deviation	data values close to mean
High standard deviation	data values widely spread

Example
Two classes have the same average score.

Class	Scores
A	60, 61, 62
B	40, 60, 80

Both mean = **60**
But class B has **larger variation**, so its standard deviation is higher.

Comparison of Measures of Variation

Measure	Meaning

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Range	difference between highest and lowest
Variance	average squared deviation from mean
Standard Deviation	average distance from mean

Final Concept Summary

Concept	Purpose
Mean	average value
Median	middle value
Mode	most frequent value
Range	spread of data
Variance	dispersion around mean
Standard Deviation	variability of dataset

Overall Data Analysis Concept

Dataset
↓
Central Tendency → Find center of data
↓
Variation → Measure spread of data

Together these measures help data scientists **summarize and understand datasets effectively**.

Measures of Shape – Skewness and Kurtosis

When analyzing data, we already studied:

- **Central Tendency** → center of data
- **Variation (Dispersion)** → spread of data

But another important property is **shape of the distribution**.

Measures of shape describe **how the data distribution looks**.

Two main measures of shape are:

Measure	Purpose
Skewness	measures asymmetry of data distribution
Kurtosis	measures the peakedness or flatness of the distribution

These help data scientists understand **how values are distributed around the mean**.

1. Skewness

Definition: Skewness measures the degree of asymmetry of a probability distribution around its mean.

If the data is perfectly symmetrical, the distribution is **not skewed**.

If one side of the distribution is longer or heavier than the other, the distribution is **skewed**.

Mathematical Formula :

skewness = (3 * (mean-median)) / standard deviation

Types of Skewness

There are **three types of skewness**.

Type	Description
Symmetrical Distribution	data evenly distributed
Positive Skewness	long tail on right side
Negative Skewness	long tail on left side

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1. Symmetrical Distribution

In a symmetrical distribution:

- left side and right side are equal
- mean = median = mode

Example:

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```

This distribution is **balanced**.

Example dataset:

4, 5, 6, 7, 8

Mean = Median = Mode

2. Positive Skewness (Right Skewed)

In positive skewness:

- tail extends toward the **right side**
- few large values pull the mean to the right

Relationship:

Measure	Value Order
Mode	smallest
Median	middle
Mean	largest

Mean > Median > Mode

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```

Example dataset:

2, 3, 4, 5, 20

Here **20** is an extreme value.

Mean becomes larger than median.

Real Example

Income distribution in many countries.

Most people earn moderate income, but **few people earn extremely high income**, creating **positive skewness**.

3. Negative Skewness (Left Skewed)

In negative skewness:

- tail extends toward the **left side**
- few small values pull the mean to the left

Relationship:

Measure	Value Order

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Mean	smallest
Median	middle
Mode	largest

```
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   /              \
```

Example dataset:
1, 15, 16, 17, 18
Here **1** is an extreme low value.
Mean < median < mode
Mean becomes smaller than median.

Summary of Skewness

Type	Shape	Relationship
Symmetrical	balanced	mean = median = mode
Positive Skew	tail right	mean > median > mode
Negative Skew	tail left	mean < median < mode

2. Kurtosis
Definition: Kurtosis measures the degree of peakedness or flatness of a data distribution compared to a normal distribution.
It describes:

- how sharp the peak is
- how heavy the tails are

Types of Kurtosis
There are **three types of kurtosis**.

Type	Description
Mesokurtic	normal distribution
Leptokurtic	sharp peak, heavy tails
Platykurtic	flat distribution

1. Mesokurtic Distribution
This distribution has **moderate peak and normal tails**.
It represents a **normal distribution**.

```
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 ---|-------|---
```

Example:
Many natural phenomena such as **human height distribution**.

2. Leptokurtic Distribution
Characteristics:

- very **sharp peak**
- **heavy tails**

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- more extreme values (outliers)

```
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```

Example:
Financial market returns often show **leptokurtic distributions**.
This means **extreme values occur more frequently**.

3. Platykurtic Distribution
Characteristics:

- **flat peak**
- **thin tails**
- fewer extreme values

```
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```

Example:
Data where values are **more evenly distributed**.

Mathematical Formula (Conceptual)
Kurtosis is often calculated using:
Kurtosis = fourth central moment / (standard deviation)^4
But in practice, software tools like:

- Python
- R
- Excel

are used to calculate kurtosis.

Comparison of Kurtosis Types

Type	Peak	Tail	Example
Mesokurtic	normal	moderate	normal distribution
Leptokurtic	sharp	heavy	financial data
Platykurtic	flat	light	evenly spread data

Importance in Data Science
Understanding skewness and kurtosis helps in:

Application	Purpose
Data distribution analysis	understand shape
Outlier detection	identify extreme values
Model selection	choose correct statistical model

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Machine learning	improve model accuracy

Many machine learning algorithms assume **normally distributed data**.

Final Concept Summary

Concept	Meaning
Skewness	measure of asymmetry
Kurtosis	measure of peakedness

Distribution Shape Overview

Skewness → direction of tail

Kurtosis → sharpness of peak

Together, they help analysts **understand the shape of data distributions and detect anomalies**.

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